XDP workshop

Netdev 0x14

August 2020

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Agenda

- **Introduction & Development updates** (5 mimutes) [*Saeed Mahameed*]
- **XDP HW hints** (15 minutes) [*Saeed Mahameed*]
- **XDP on hosts pain points summary** (5 minutes) [*David Ahern*]
- **XDP egress, development status and use case** (10 minutes) [*David Ahern*]
- **Multi-buffer for XDP** (10 minutes) [*Sameeh Jubran*]
- **XDP for TSN(time sensitive networking) and XDP time-stamping** (10 minutes) [*Ilias Apalodimas*]
- **Busy poll support for AF_XDP sockets** (10 minutes) [*Sridhar Samudrala*]
- Open mic for discussion & concluding remarks, 6:20 - 6:45pm (25 minutes)
  - Maintaining XDP performance [*Jesper*]
  - Page pool skb recycling & iommu - [*Jonathan Lemon*]
  - 0-copy with header data split - [*Jonathan Lemon*]
  - Open discussions

- *Will allow 5 minutes Q/A and free discussion after each topic*

XDP Development updates

- XDP programs in CPUMAP - [*Lorenzo Bianconi*] [1]
- XDP programs in DEVMAP (XDP Egress) - [*David Ahern*] [2]
- bpf_xdp_adjust_tail() to grow packet size at tail - [*Jesper Dangaard Brouer*] [3]
- bulking for non-map XDP_REDIRECT, +68% packet rate - [*Toke Høiland-Jørgensen*] [4]
- Bulk optimization for XDP cpumap redirect - [*Jesper Dangaard Brouer*] [5]
- bpf: Introduce dynamic program extensions - [*Alexei Starovoitovc*] [6]
- Numa aware page pool - [*Saeed Mahameed & Jonathan Lemon*] [7]

AF_XDP:

- add need_wakeup flag to the AF_XDP rings - [*Magnus Karlsson*] [8]
- Introduce AF_XDP buffer allocation API – [*Björn Töpel & Maxim Mikityanskiy*] [9]
- xsk: add new statistics - [*Ciara Loftus*] [10]

Welcome to the club 2019-2020

- Mvpp2, mvneta : Marvell's Armada SoC
- ena: Amazon's Elastic Network Adapter (ENA) driver
- netsec: SocioNext NetSec Gigabit Ethernet
- sfc: Solarflare's SFC 10/40-gigabit Ethernet cards.
- xen-netfront: xen frontend network driver
- hv_netvsc: Hyper-V virtual network driver

Closing the gap:

- Support XDP_REDIRECT in qede driver
- Mlx5 XSK support
- veth: Support bulk XDP_TX
- dpaa2-eth: add XDP_REDIRECT support

In-Progress

- XDP metadata HW hints – [*Saeed Mahameed*]
- XDP Multi buffer – [*Sameeh Jubran*]
- net/bridge: access FDB from XDP programs - [*Yoshiki Komachi*]
- NetGPU: GPU zero copy from XDP rings – [*Jonathan Lemon*]

- Please speak up today if you are working on something interesting :-)

XDP HW hints

Saeed Mahameed

XDP pain points & XDP Egress

David Ahern

XDP Multi-Buffer

Sameeh Jubran

XDP for TSN – By *Ilias Apalodimas*

- TSN aims to provide bounded latency on Ethernet based LANs.
- Low latency is (probably) an issue. There are efforts trying to reduce the bounded
  worst-case latency of the Linux network stack: AF_XDP.
- Initial measurements on Arm devices show ~70-100us for the default network stack and sub 20us for AF_XDP
- Merged schedulers in kernel (ETF, TAPRIO)
- XDP timestamping

- https://elinux.org/images/5/56/ELC-2018-USA-TSNonLinux.pdf

Busy-poll support

Sridhar Samudrala

Performance & Page pool

- Performance gains and usage:
  - https://s3.amazonaws.com/connect.linaro.org/bkk19/presentations/bkk19-505.pdf
- Initial RFC: https://lore.kernel.org/netdev/154413868810.21735.572808840657728172.stgit@firesoul/

Changes since RFC:

- Added a pp_recycle bit in the SKB to trigger recycling
 - All of the recycle info are kept on struct page. This allows us to recycle SKB fragments.
 - Jonathan added protections with extra checks on the page, to make sure we recycle page pool buffers (no github tree for these)

Problems:
 - Split page memory model won't work with the current code (Intel, Mellanox)
   Is it possible to recycle the page once both fragments are used? i.e propose an page_pool type?

Open Discussion

- Maintaining XDP performance
  - Micro benchmarks? Self-tests ?
- Page pool, skb recycling & iommu
- 0-copy with header data split
- Enabling TX offloads ? Using BTF meta data ?
- Networking Queues: (XDP TX, AF_XDP) https://linuxplumbersconf.org/event/4/contributions/462/attachments/241/422/LPC_2019_kernel_queue_manager.pdf
- Unifying XDP statistics reporting and accounting
- More ...

References

- [1] https://lwn.net/ml/bpf/cover.1594734381.git.lorenzo@kernel.org/
- [2] https://lwn.net/ml/netdev/20200528001423.58575-1-dsahern@kernel.org/
- [3] https://www.spinics.net/lists/bpf/msg17519.html
- [4] https://lwn.net/Articles/809796/
- [5] https://patchwork.ozlabs.org/project/netdev/cover/155489659290.20826.11087703475112926 18.stgit@firesoul/
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